

IN-HOUSE SCHEDULE

9:00 – 9:30

OPENING REMARKS

Location: **LSC 4260**

9:00 – 9:10

Aaron Newman

Opening Remarks

9:10 – 9:25

Ray Klein

50+ Years in Psychology (and Neuroscience) at Dalhousie

9:30 – 10:30

SESSION A | History, Personality, and People

Session Chair: **TBD**

Location: **LSC 4260**

A-1

9:30 – 9:45

Erin Sparks

TBD

A-2

9:45 – 9:55

Julie Blais

Personalized Persuasion? Testing the Efficacy of Personality-Targeted Political Advertisements

A-3

9:55 – 10:05

Isabella Bossom

Explaining Unique and Common Variance in Financially Focused Self-Concept from Dark Triad and Honesty-Humility Personality Traits Using Commonality Analyses

A-4

10:05 – 10:15

Christopher Park

The Emotional Demands of Civilian Crisis Work: Exploring Burnout, Role Clarity, and Support in Civilian Crisis Intervention Teams

A-5

10:15 – 10:25

Sasha Stephen

Experimental Manipulations to Improve Participant Comprehension of Informed Consent Forms

10:30 – 11:00

COFFEE BREAK

Location: **LSC 4332** (Lilyan E. White Study Centre)

Please return to LSC 4260 at 10:55

11:00 – 12:00

SESSION B | Attention and Computational Approaches

Session Chair: **TBD**

Location: **LSC 4260**

B-1 11:00 – 11:10

Jeremy Webster

Experimental Investigations of Smartphone Use and Attention

B-2 11:10 – 11:20

Reyhana Wilson

The Effects of Acute Short-Form Video Viewing on Go/No-Go Performance

B-3 11:20 – 11:30

Ralph Redden

From Bonding to Focus: Examining the Relationship Between Attachment Styles and Networks of Attention

B-4 11:30 – 11:40

Nicholas Murray

Assessing Attentional Bias for Cannabis Activated by Trauma Cue Exposure in Cannabis Users with Trauma Histories

B-5 11:40 – 11:50

Ryan MacArthur

Developing a Deep-Learning Model for Individual Recognition in Canadian Geese: Extending Avian Observational Research Through Computer Vision

B-6 11:50 – 12:00

Can Sozuer

Quantitative Data-Based-Evaluation of Signal-to-Noise in Assessments Reveals Novel Insights

12:00 – 13:30

LUNCH AND POSTER SESSION

Location: **LSC 4332** (Lilyan E. White Study Centre)

Please return to LSC 4260 at 13:25

13:30 – 14:30

SESSION C | Neuroscience, Vision, and Language

Session Chair: **TBD**

Location: **LSC 4260**

C-1 13:30 – 13:45

Richard E. Brown

The Influence of Ramon y Cajal and Rafael Lorente de Nó on Donald O. Hebb

C-2 13:45 – 13:55

Andrew Butler

Determining Cortical Roles in Motion Perception via Optogenetic Inactivation of Visual Areas

- C-3** **13:55 – 14:05** **Jeffrey Locke**
Seeing More in the Electroretinogram: Integrating Time, Frequency, and Resting-State Analyses Across Mouse Models and Human Retinal Disease
- C-4** **14:05 – 14:15** **Hope Lewis**
How Quickly Does Neural Activity Change in Response to Learning Unfamiliar Language Vocabulary?
- C-5** **14:15 – 14:25** **Shuqi Yu**
One Skill, Different Journeys? Morphological Awareness and Reading Growth Across Socioeconomic Status

14:30 – 15:00 **COFFEE BREAK**
Location: **LSC 4332** (Lilyan E. White Study Centre)
Please return to LSC 4260 at 14:55

15:00 – 16:00 **SESSION D | Memory, Stress, and Health**
Session Chair: **TBD**
Location: **LSC 4260**

- D-1** **15:00 – 15:10** **Ian Dauphinee**
Semantic Similarity Disrupts Order Recall: The Moderating Role of Task Difficulty
- D-2** **15:10 – 15:20** **Mackenzie R. Hartley**
The Persistent Effects of Adolescent Chronic Psychological Stress on Sociability in Male and Female Mice
- D-3** **15:20 – 15:30** **Megan Babcock**
Expanding on the Two-Hit Model of Depression-Like Behaviour and the Underlying Role of the Oxytocin System
- D-4** **15:30 – 15:40** **Caden Whitfield**
The Impact of Income Level on Child Sleep, Parental Engagement, and Perceived Effectiveness of the Better Nights, Better Days eHealth Intervention
- D-5** **15:40 – 15:50** **Mikaela Ethier-Gagnon**
The Impact of Menstrual Cycle Phase on Subjective, Behavioural, and Physiological Responses to Electronic Cigarette Ingredients
- D-6** **15:50 – 16:00** **Elena Koning**
Psilocybin Treatment as an Adjunct to Cognitive Behavioral Therapy for Eating Disorders: Therapeutic Rationale & Considerations for Protocol Development

SESSION A TALK ABSTRACTS

A-1 **Erin Sparks**

TBD

TBD

A-2 **Julie Blais**, Scott Pruyers, and Luke Mungall

Personalized Persuasion? Testing the Efficacy of Personality-Targeted Political Advertisements

Before we can evaluate the harm that psychographic microtargeting may entail, we must first empirically examine whether political advertising can be tailored to resonate with people at the level of their personality. In Study 1, we conducted a survey among personality experts ($N = 86$) to evaluate the face validity of real-world targeted ads used for a US midterm election. In Study 2, we conducted a pre-registered between-subjects experimental design with a nationally representative sample of over 2,000 US citizens to assess whether advertisements that match participants' personality traits are more effective (i.e., vote intention, likability, etc.). Study 1 indicated that a majority (63%) of experts in personality psychology were unable to correctly identify the Big Five trait that was targeted in the video. In Study 2, there were no significant aggregate effects of the matching variable, once controlling for participant personality. However, a classification tree analysis did identify a small sub-group of participants where matched ads had higher efficacy. While concerns about psychological voter manipulation are widespread, the results of these studies suggest that such fears are likely overstated. However, even small and subtle effects when scaled to tens of millions of voters may influence election outcomes.

A-3 **Isabella Bossom**, Sarah Enouy, Julie Blais, Michael Wohl, Scott Pruysers, and Nassim Tabri

Explaining Unique and Common Variance in Financially Focused Self-Concept from Dark Triad and Honesty-Humility Personality Traits Using Commonality Analyses

We investigated the personality correlates of financially focused self-concept, a construct describing individuals who overemphasize financial success as central to their self-definition and self-worth. Although some of the consequences of a financially focused self-concept have been studied, little is known about the personality characteristic of financially focused people. To address this gap, we examined the unique and shared associations between financially focused self-concept and the Dark Triad traits (Machiavellianism, psychopathy, and narcissism) as well as Honesty-Humility at the facet level to provide greater specificity than trait-level analysis. Using cross-sectional data from 1,725 Canadian adults who completed validated self-report measures, we conducted commonality analyses. Results revealed significant associations between financially focused self-concept and Machiavellianism facets (antagonism, planfulness), all psychopathy facets, narcissistic vulnerability, and lower levels of Honesty-Humility (greed-avoidance and modesty). These findings enhance understanding of the personality profiles of individuals who prioritize financial success excessively.

A-4 **Christopher Park**, Chris Giacomantonio, and Lori Wozney

The Emotional Demands of Civilian Crisis Work: Exploring Burnout, Role Clarity, and Support in Civilian Crisis Intervention Teams

Civilian Crisis Intervention Teams (CCITs) are rapidly expanding as alternatives to police-led mental health crisis response across Canada and the United States, yet little is known about responder well-being. Grounded in the Job Demands-Resources (JD-R) framework, this study examined burnout, perceived organizational support (POS), and role clarity during police collaboration among CCIT responders. Participants (N = 50) were current or recent civilian crisis responders who completed an anonymous online survey including the Copenhagen Burnout Inventory, the Perceived Organizational Support Scale-10, the Brief Role Clarity in Police Collaboration Scale, and demographic items. Descriptive statistics, reliability analyses, Pearson correlations, and hierarchical multiple regression were conducted. Responders reported moderate-to-high burnout, with the highest levels in personal and work-related domains. POS was significantly and negatively associated with burnout and remained a significant predictor in a model explaining 28.6% of burnout variance, even after accounting for years of experience. In contrast, role clarity did not significantly predict burnout, despite being positively associated with POS. These findings suggest POS may be a central factor in understanding burnout among CCIT responders. As CCIT's continue to expand, strengthening POS may be critical for sustaining CCIT models.

Experimental Manipulations to Improve Participant Comprehension of Informed Consent Forms

Though informed consent is a critical part of research ethics, research suggests that comprehension is low. The present study explores methods to improve the comprehension of consent forms via supplementary quizzes and videos. Using an online survey, undergraduates ($n = 533$) were given a standard consent form. Each participant was randomized to one of 4 conditions: (a) a control condition with standard consent; (b) adding a practice quiz; (c) watching a summary video with a male or (d) female actor. The primary outcome was scores on a multiple-choice quiz of facts related to the consent form. A secondary outcome was whether participants chose to share their data in a repository. Both video conditions were superior to control, with the male video condition having the best comprehension. The quiz condition was not statistically different from control. Though comprehension on most questions was good (~70-80%), understanding of open-access repositories was generally poor (~40%). Despite poor understanding, data sharing in repositories was high (~94%) and there were no differences by condition. Using video summaries following a written consent form can improve comprehension of the consent form's content. However, more work needs to be done to improve participants comprehension of broad consent.

SESSION B TALK ABSTRACTS

B-1 **Jeremy Webster** and Raymond Klein

Experimental Investigations of Smartphone Use and Attention

A review of research and literature on the interaction between smartphone use and attention. Includes an analysis of present limitations in the field when aiming to assess causality. Review of most commonly practiced experimental designs on the subject: notification, mere presence (brain drain), acute use, prolonged use. Results are interpreted using a contemporary taxonomy of attention (Klein, 2022), and suggest a negative influence on endogenously allocated attention and executive control. Consideration of future directions concludes the review.

B-2 **Reyhana Wilson**, Heather Neyedli, and Cory Munroe

The Effects of Acute Short-Form Video Viewing on Go/No-Go Performance

Short-form videos (SFVs), typically lasting from a few seconds to a few minutes, have become increasingly popular across social media platforms. Although SFV use has been associated with poorer cognition, particularly inhibitory control and attention (Nguyen et al., 2025), experimental evidence testing its acute effects remains limited. This study examined whether brief SFV viewing, relative to a reading control, impairs Go/No-Go task performance, which requires response inhibition and sustained attention. Using a counterbalanced crossover design, participants ($N = 36$, $M_{age} = 18.6$, $SD = 1.38$) completed two 15-minute smartphone-based conditions: SFV viewing and reading. Immediately after each condition, participants completed a Go/No-Go task and self-report questionnaires. Preliminary analyses showed that participants reported an average of 31.8 hours ($SD = 15.05$) of smartphone use per week, with 44% of this time spent viewing SFVs ($M = 14.07$, $SD = 9.98$). The most viewed SFV genres included entertainment, music, people/blogs, comedy and news/politics. Our sample reported substantial SFV viewing, highlighting the popularity of this form of media. The effects of acute SFV viewing on Go/No-Go performance will also be presented.

B-3 Kyra Battist, Veda Haggarty, Julia Byron, and **Ralph Redden**

From Bonding to Focus: Examining the Relationship Between Attachment Styles and Networks of Attention

Relationship attachment style has been found to influence performance associated with two of the three networks of attention. Attachment avoidance has been shown to influence the executive network of attention, wherein individuals higher in attachment avoidance show reduced interference in the standard Eriksen Flanker task. Attachment anxiety has been shown to influence the executive network as well, in addition to the orienting network of attention, wherein individuals higher in attachment anxiety show larger endogenous cueing effects. The current study was designed to replicate these prior findings, and to examine the heretofore untested relationship between attachment styles and the alerting network of attention, all in a single task. AttentionTrip is an engaging, gamified version of the Attention Network Task, played on an iPad tablet. Attachment styles were assessed using the Experience in Close Relationships-Revised Questionnaire. Robust network scores were observed for each network of attention. Attachment styles were associated with inverse relationships with overall RT. Moreover, orienting and executive control interacted differently with both attachment styles, whereas neither attachment style reliably influenced the alerting network. These findings support the hypothesis that the two forms of insecure attachment are associated with different underlying attentional functioning.

B-4 **Nicholas Murray**, Sarah DeGrace, Melissa Burke, Isabella Bossom, Mikaela Ethier-Gagnon, Raymond Klein, and Sherry Stewart

Assessing Attentional Bias for Cannabis Activated by Trauma Cue Exposure in Cannabis Users with Trauma Histories

Posttraumatic stress disorder (PTSD) and cannabis use disorder (CUD) co-occur at rates five times higher than chance. An integrated self-medication/attentional bias (iSM-AB) model suggests that trauma reminders may shift attention toward cannabis cues in cannabis users with trauma histories, increasing cravings and intentions to use. This project tests the iSM-AB model using a personalized cue reactivity paradigm and an Addiction Stroop Task. To date, 17 (aiming to continue recruitment for a total of 50) participants completed both sessions. In Session 1, participants completed questionnaires and a semi-structured interview to generate personalized trauma and neutral scripts. In Session 2, participants were exposed to trauma and neutral cues, each followed by an Addiction Stroop Task with cannabis-related and neutral words. Preliminary linear mixed model results indicate that among individuals with higher PTSD symptoms, trauma cue exposure (but not neutral exposure) increases cannabis-related attentional bias ($\beta = 0.43$, $p = .04$). Post-cue negative affect was also positively associated with attentional bias, but only in the trauma condition ($r = 0.46$, $p = .03$). These preliminary findings support the iSM-AB model and suggest attentional bias modification training may be best targeted to individuals with more severe PTSD symptoms when preceded by trauma cue exposure.

B-5 Ryan MacArthur

Developing a Deep-Learning Model for Individual Recognition in Canadian Geese: Extending Avian Observational Research Through Computer Vision

Individual recognition is a foundational cognitive pillar that supports complex social architectures across the animal kingdom, enabling behaviours such as long-term pair bonding, reciprocal altruism, and hierarchy maintenance. Studying these behaviours requires both establishing cognitive capacity and developing non-invasive tools to observe undisturbed behaviour. Avian ethology has emphasized auditory recognition, yet mounting evidence shows visual signals also carry identity information. Work on the Greylag Goose (*Anser anser*) has demonstrated bill- and face-level individuality with high algorithmic accuracy. The Canadian Goose (*Branta canadensis*) presents as a similar candidate and remains unexplored. Here we report a computational study comparing two deep-learning architectures trained and tested on identical data, as a step toward non-invasive individual identification. Individually-labelled images were assembled from a reference dataset and from video-derived exports produced by a purpose-built annotation tool. We tuned a ResNet-50 classifier (closed-set assignment) and a MegaDescriptor-L-384 retrieval model (gallery matching). Both reached >90% accuracy on novel test images. Saliency maps emphasized the cheek-patch boundary, bill, and facial regions as candidate identity cues. Together, the results support a passive, camera-based alternative for avian observational research.

B-6 Can Sozuer

Quantitative Data-Based-Evaluation of Signal-to-Noise in Assessments Reveals Novel Insights

Online learning platforms like Brightspace, Remark, or Crowdmark can display evaluative metrics for questions graded on their platforms. These metrics estimate how well a question assesses student learning by comparing how well performance on that question predicts performance on other questions (signal). However, platform-provided metrics may not evaluate well assessments with a wide variety of question types, as this variation reduces predictiveness. They also cannot incorporate question information supplied externally by a teaching user to account for between question variation. Here, I present a new metric for evaluating how well a question assesses learning, with the added ability to incorporate externally provided question information. This metric is thought to evaluate how well a question assesses learning, especially on high-variation assessments, more accurately than readily available alternatives. I then apply this metric to datasets to extract insights into which assessment design elements increase how well a question assesses student learning (signal).

SESSION C TALK ABSTRACTS

C-1 **Richard E. Brown**

The Influence of Ramon y Cajal and Rafael Lorente de Nó on Donald O. Hebb

Donald O. Hebb obtained his BA in English and Philosophy from Dalhousie University in 1925 and his MA in Psychology from McGill in 1932. He completed his PhD with Karl Lashley at Harvard in 1936. Between 1932 and 1934 he wrote three unpublished papers on the neural basis of behaviour. After his PhD he conducted neuropsychological tests on the patients of Dr. Wilder Penfield at the Montreal Neurological Institute (1937-39). From 1939 to 1942 he was a professor at Queen's University in Ontario and then he studied emotionality in chimpanzees at the Yerkes Primate Centre in Florida (1942-1947). It was here in 1944 that he began to revise his old theory of the neural basis of behaviour and wrote five sets of unpublished notes for his influential book, *The Organization of Behavior*. This book introduced the concepts of the "Hebb synapse" and learning rule, cell assemblies, and phase sequences to account for the neural events underlying behaviour. The crucial element in Hebb's theory was the reverberatory circuit of Lorente de Nó, based on Cajal's concept of closed circuits. This presentation examines the unpublished notes and papers written by Hebb as he developed his neuropsychological theory.

C-2 **Andrew Butler**, Nicole Michaud, Lan Dang, and Nathan Crowder

Determining Cortical Roles in Motion Perception via Optogenetic Inactivation of Visual Areas

Motion perception is an integrative process that combines information about the direction, size, and velocity of a stimulus. Our aim is to examine the cortical networks underlying speed discrimination, which is the ability to perceive differences in velocity. The optogenetic regulation of an inhibitory interneuron circuit in the mouse cortex allows for reversible inactivation of various cortical areas to determine which areas are involved with speed discrimination. By combining this optogenetic technique with a 2-alternative forced-choice task where the mouse must select the faster of two random-dot kinematograms, we are able to assess if the targeted region is involved in speed discrimination. Pilot data has already been collected showing that optogenetic inactivation of the primary visual cortex (V1) can impact motion perception. From here we plan to conduct a more complete experiment to fully assess the role of V1 and higher visual areas in speed discrimination. This research provides foundational information about the contributions of individual cortical regions to visual perception, but also provides a methodological framework for future research to assess the relevance of individual cortical areas to any behaviourally measurable aspect of perception or cognition due to canonical circuitry.

C-3 Jeffrey Locke, Patrice Côté, and Aaron Newman

Seeing More in the Electroretinogram: Integrating Time, Frequency, and Resting-State Analyses Across Mouse Models and Human Retinal Disease

Interpretation of the electroretinogram (ERG) has traditionally focused on time-domain waveform features which may oversimplify complex retinal signaling. Frequency- and time-frequency-domain analyses offer complementary perspectives but remain inconsistently applied due to methodological variability. Resting-state ERG has also emerged as a potential probe for intrinsic retinal activity, though its relationship to evoked responses remains poorly defined. Full-field ERGs were recorded across stimulus-evoked and resting-state conditions in mouse models representing normal retina (n=23), inner-retinal dysfunction (n=10), and global-retinal degeneration (n=11). Signals were analyzed using time-domain metrics, power spectral density, and time-frequency transforms. Time-frequency analysis employed Morlet wavelets parameterized using a full-width-at-half-maximum (FWHM) framework to explicitly control temporal-spectral trade-offs. An identical analytical pipeline was applied to human clinical ERG and resting-state data (n=69). Progressive retinal dysfunction produced graded reductions in waveform amplitude, spectral power, and time-frequency energy, with broad, permutation-corrected spectral and temporal clusters distinguishing phenotype. Resting-state ERG demonstrated consistent alterations in aperiodic spectral slope and high-frequency structure, persisting even when evoked responses approached the noise floor. Comparable patterns were observed across species. Integrating time-domain, frequency-domain, time-frequency, and resting-state analyses provides a more sensitive and comprehensive assessment of retinal function than any signal method alone.

C-4 Hope Lewis, Cate Brandt, and Aaron Newman

How Quickly Does Neural Activity Change in Response to Learning Unfamiliar Language Vocabulary?

Semantic processing is associated with the N400 event-related potential (ERP), with larger N400 amplitudes appearing when information mismatches versus when it matches expectation. This differentiation is called the N400 effect, which does not appear with unfamiliar language (UL) vocabulary before training, but does appear by 10 days after training begins. By recording EEG throughout the training process, the current study determined how much exposure to UL vocabulary a new learner needs to show an N400 effect in response to said UL vocabulary. Over two sessions, Spanish vocabulary was taught to UL learners by pairing words with images representing their meaning. A semantic mismatch task paradigm was used to assess N400 ERPs in Spanish and in English before and after each block of training in each session. We found that after 3-4 exposures to UL words in the first session, an N400 effect emerged, with this remaining during the second session. Mean amplitude was smaller in the Spanish vocabulary compared to English, reflective of the difference in proficiency. Accuracy of responses also improved over time, reaching ceiling by the end of day 1. Findings provide useful insight into the earliest neural markers of acquiring new semantic information with UL vocabulary.

One Skill, Different Journeys? Morphological Awareness and Reading Growth Across Socioeconomic Status

Reading comprehension supports learning across academic subjects. A robust predictor is morphological awareness (MA), the ability to recognize and manipulate meaningful word units. Although higher-SES children show faster reading growth especially in Grade 1, it remains unclear how SES shapes the long-term impact of Grade 1 MA on reading growth. Therefore, this study examined whether MA contributes differently across socioeconomic contexts from Grade 1 to Grade 4. Participants were 313 English-speaking children from 18 primary schools in Eastern North America. In Grade 1, children completed MA, reading comprehension, and control measures (word reading, vocabulary, phonological awareness, nonverbal ability, working memory). Parents reported family demographics, including SES. In Grade 4, standardized reading comprehension was assessed. Grade 1 MA uniquely predicted Grade 4 reading comprehension ($\beta = 0.51, p < .001$). SES was also positively associated with later reading comprehension ($\beta = 0.10, p = .01$). A significant MA $\times$ SES interaction indicated that the association between MA and reading comprehension growth was stronger for children from lower-SES backgrounds ($\beta = -0.11, p = .02$). These findings suggest that MA is particularly important for reading development among lower-SES children, highlighting its potential as a target for early instruction.

SESSION D TALK ABSTRACTS

D-1 **Ian Dauphinee**, Dominic Guitard, Sho Ishiguro, and Jean Saint-Aubin

Semantic Similarity Disrupts Order Recall: The Moderating Role of Task Difficulty

Similarity affects memory across many stimulus domains, typically impairing short-term order recall, with one important exception: semantically similar words (e.g., apple, orange, banana). In fact, the challenging insensitivity of short-term order recall performance to semantic similarity has become a benchmark effect for evaluating memory models. However, contrary to the literature, three key studies reported a detrimental effect of semantic similarity on short-term order recall. Here, in five large-scale experiments, we revisited these three exceptions. Despite implementing more rigorous methodological controls, we reproduced the detrimental effect of semantic similarity across all experiments. We then conducted a review and meta-analysis, which revealed a small but reliable detrimental effect across the literature. To account for previous inconsistencies, we identified a potential moderating variable: task difficulty. Across four additional experiments, we manipulated task difficulty via presentation speed and list length. As predicted, under more difficult task conditions semantic similarity impeded order recall, and critically, when the task was easier, its detrimental effect on order recall vanished. These findings show that semantic similarity is not a special case for short-term order memory and that its effect depends in part on task difficulty, helping reconcile previously conflicting results for models of serial order memory.

D-2 **Mackenzie R. Hartley** and Tamara B. Franklin

The Persistent Effects of Adolescent Chronic Psychological Stress on Sociability in Male and Female Mice

Adolescents are more sensitive to stress than adults and experiencing stress during this developmental period can have lasting effects on behaviour, including increased depression-like and anxiety-like behaviours. Our research aims to determine the effects of chronic adolescent stress on adult sociability, and the neural mechanisms that may underlie these effects. Our behavioural research has found that adolescent Chronic Predator Stress (aCPS) exposure leads to an increase in same-sex peer investigation and altered social dominance behaviours in adult mice. Gene expression in the brains of these animals was analysed using rt-qPCR analyses, with an initial focus on the dorsal and ventral hippocampus, brain regions that are still developing during adolescence, are responsive to stress, and have been shown to be critical for social function. We quantified *Drd2* and *Oxtr* (genes that encode for dopamine and oxytocin receptors), and histone deacetylase 2 (*Hdac2*), all of which represent systems that have been associated with both social behaviours and stress. aCPS had no effect on hippocampal expression of these genes in male or female mice. Future work includes a brain-wide c-Fos mapping study to gain insight into the brain areas likely to contribute to the altered peer investigation behaviours observed in our behavioural study.

D-3 Megan Babcock, Kate Zucconi, Janet Menard, and Tamara Franklin

Expanding on the Two-Hit Model of Depression-Like Behaviour and the Underlying Role of the Oxytocin System

The two-hit hypothesis proposes that early-life stress increases vulnerability to later stressors. While this hypothesis was originally developed in schizophrenia research, recent work has applied it to depression as well. My MSc research tested whether early-life adversity (ELA) heightens sensitivity to subthreshold stress. Rats exposed to both neonatal limited bedding and nesting stress and a subthreshold version of chronic mild stress in adolescence displayed greater depression-like behaviour and aggression in adulthood, whereas neither stressor produced effects alone. LBN exposure was also associated with reduced oxytocin (OT)-immunoreactive cells in the paraventricular nucleus, a key OT-producing region that helps regulate stress responses and social behaviour. These findings suggest that ELA renders individuals more vulnerable to depression-like behaviour following a later stressor, even when subthreshold, and implicate altered OT signaling as a potential underlying mechanism. Building on this work, my proposed PhD research will test the hypothesis that ELA disrupts OT system function, and that these disruptions have lasting consequences for stress sensitivity across development. Specifically, I will investigate whether the timing and type of stress exposure differentially alter OT production, release, and receptor signaling, and whether these changes drive depression-like and social behavioural outcomes.

D-4 Caden Whitfield, Mya Dockrill, Penny Corkum, Debbie Emberly, Christine Cassidy, Alissa Pencer, and Guido Simonelli

The Impact of Income Level on Child Sleep, Parental Engagement, and Perceived Effectiveness of the Better Nights, Better Days eHealth Intervention

Behavioural sleep disturbances (e.g., insomnia) are common in school-aged children and negatively impact health and functioning. *Better Nights, Better Days (BNBD)* is an evidence-based eHealth intervention proven to improve sleep in children. Given that most intervention research mainly includes participants with middle to high income, we do not know how income level impacts children's sleep and treatment for sleep problems. A secondary analysis of four previous *BNBD* studies was performed to examine differences between families with low income ($n = 145$) and mid-high income ($n = 820$) regarding sleep problem severity, engagement with the intervention, and perceived effectiveness. Data from sleep health questionnaires demonstrated that low-income children had more severe sleep problems. User analytics (i.e., number of sessions completed) from the *BNBD* intervention showed very similar rates of engagement between income-level groups. Differences between pre- and post-intervention questionnaires for participants who completed a "clinical dose" of the program showed that the low-income participants reported lower effectiveness than the mid-high-income participants. This study supports that sleep health disparities are present in Canadian children and demonstrates the promise of *BNBD* as a usable intervention for low-income families, while emphasizing the need to further investigate the intervention's effectiveness for low-income families.

D-5 **Mikaela Ethier-Gagnon**, Makayla MacLean, Robin Perry, Cory Munroe, Tara Perrot, Sherry Stewart, Kim Good, and Sean Barrett

The Impact of Menstrual Cycle Phase on Subjective, Behavioural, and Physiological Responses to Electronic Cigarette Ingredients

Electronic nicotine delivery systems (ENDS) use has increased in recent years, partly due to appealing flavours such as menthol. Sensory factors appear particularly important for nicotine reinforcement in females, and ovarian hormones may modulate these effects. However, the role of the menstrual cycle in menthol- and nicotine-related reinforcement remains largely unexamined. Naturally cycling female ENDS users (N = 34) completed experimental sessions during two menstrual phases (follicular and luteal phase) in a randomized mixed within-subjects (menstrual phase) and between-group (menthol vs. menthol-free) design. Immediately before and after a standardized puffing procedure, subjective measures (craving, withdrawal, subjective state, inhaler effects) and heart rate variability (HRV) were assessed. Salivary estradiol and progesterone were collected to validate menstrual phase and for exploratory analyses. Menthol enhanced sensory and reward-related subjective effects and was associated with greater reductions in relief craving relative to the menthol-free condition. In the menthol-free condition, participants in the follicular phase reported greater withdrawal and subjective 'high' relative to the luteal phase. Exploratory analyses indicated that higher progesterone levels were associated with reduced withdrawal in the menthol-free condition only. Results support policy efforts regulating menthol in ENDS products and highlight the need for targeted cessation strategies for female users.

D-6 **Elena Koning**, Susan Gamberg, and Aaron Keshen

Psilocybin Treatment as an Adjunct to Cognitive Behavioral Therapy for Eating Disorders: Therapeutic Rationale & Considerations for Protocol Development

Eating disorders (EDs) remain challenging to treat, with high dropout and low remission rates in cognitive-behavioral therapy for EDs (CBT-ED). Psilocybin treatment (PT) demonstrates therapeutic potential to enhance CBT-ED by exerting several neurobiological (e.g., neuroplasticity, cognitive flexibility), psychological (e.g., antidepressant), and experiential (e.g., illness insights) effects. Specifically, the effects of PT are hypothesized to increase psychotherapeutic engagement/learning, reduce dropout, and improve clinical outcomes. This perspective article (recently published in Behavioral Sciences) provides the first consolidation of neurobiological and transdiagnostic clinical evidence for PT/CBT-ED, proposes considerations for a concurrent intervention protocol, and presents clinical research considerations to empirically test its feasibility, safety, and efficacy. This line of inquiry is expected to advance the development of approaches that improve ED treatment outcomes. More broadly, it advances the study of psychedelics as tools to enhance evidence-based psychotherapy models for difficult-to-treat neuropsychiatric conditions.